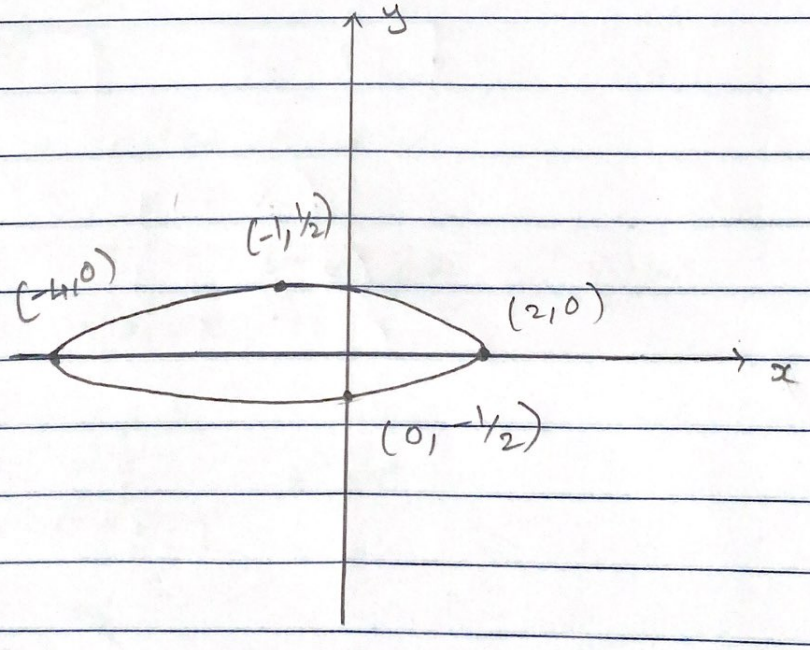
Quiz 17:-

(16): $x = 3\sin(t) - 1$ and $y = \frac{\cos t}{2}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-\sin t/2}{3\cos t} = \left(-\frac{2}{3}\right)(\tan t)$$

t	$x = 3\sin(t) - 1$	$y = \frac{\cos(t)}{2}$	(x, y)
0	-1	$1/2$	$(-1, 1/2)$
$\pi/2$	2	0	$(2, 0)$
π	0	$-1/2$	$(0, -1/2)$
$3\pi/2$	-4	0	$(-4, 0)$
2π	-1	$1/2$	$(-1, 1/2)$



C:17 $x = t+3$ and $y = 1-t^2$ at $t=1$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\frac{dy}{dx} = \frac{-2t}{1}$$

$t=1 \Rightarrow \frac{dy}{dx} = -2$

$y - y_1 = m(x - x_1)$

$y_1 = 1-1 = 0$, $x_1 = 1+3 = 4$ $(4, 0)$

$\Rightarrow y - 0 = -2(x-4)$

$\Rightarrow \boxed{y = -2x + 8}$

C18:- $\tan\theta = y/x$; $x^2 + y^2 = r^2$, $x = r\cos\theta$, $y = r\sin\theta$

(A) $\tan\theta = \frac{-3\sqrt{3}}{-3}$

$\tan\theta = \sqrt{3} \Rightarrow \theta = \tan^{-1}(\sqrt{3})$

$\boxed{\theta = \pi/3}$

$(r\cos(\pi/3))^2 + (r\sin(\pi/3))^2 = r^2$

$\Rightarrow r^2\left(\frac{1}{4}\right) + r^2\left(\frac{3}{4}\right) = r^2$

$\Rightarrow \frac{1}{4} + \frac{3}{4} = 1$

$\Rightarrow 1 = 1$

$\boxed{x^2 + y^2 = r^2}$

$\Rightarrow (-3)^2 + (-3\sqrt{3})^2 = r^2$

$\Rightarrow 9 + 27 = r^2 \Rightarrow r = \pm\sqrt{36}$

$\Rightarrow \boxed{r = \pm 6}$

$\boxed{\theta = \pi/3}$ $\boxed{r = \pm 6}$



$(r, \theta) = (6, \pi/3)$

$(r, \theta) = (-6, 2\pi/3)$

$(r, \theta) = (-6, 4\pi/3)$

(B) $r^2 \sin(2\theta) = 1$